SYSTEM DESIGN DOCUMENT

DealerDesk

Dealership inventory: high-level and low-level design

Part 1 · High-level design

System context, deployment on Cloudflare and Neon, edge routing, the user journey, and the CI/CD pipeline.

Part 2 · Low-level design

Backend layers, login, token validation, the atomic vehicle update, concurrency, the frontend edit flow, the data model and the API contract.

Live application	https://dealerdesk.aneeq02025.workers.dev
Repository	github.com/aneeq178/dealership-inventory
Stack	Next.js 16 · FastAPI on Cloudflare Python Workers · PostgreSQL (Neon) via Hyperdrive · GitHub Actions
Generated	2026-10-25 from docs/ARCHITECTURE.md

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How to read this document

What high-level design means

High-level design (HLD) describes the system from the outside. It answers:

- **What are the major parts?** For example: a web frontend, an API, a database, a delivery pipeline.
- **Where does each part run?** For example: Cloudflare, Neon, GitHub.
- **How do the parts talk to each other?** Which protocols, and in which direction.
- **Who uses the system, and for what?**

HLD deliberately leaves out function names, SQL and field types. Its audience is anyone who needs the big picture: reviewers, stakeholders, a new team member on day one.

What low-level design means

Low-level design (LLD) opens each box from the HLD and describes how it works inside. It answers:

- **How is the code organised?** Modules, layers, and their responsibilities.
- **Exactly what happens, step by step,** when a request arrives: every check, every decision, every database call.
- **What are the data structures?** Tables, columns, constraints, API contracts and error codes.
- **How are the hard cases handled?** Invalid tokens, wrong roles, other dealerships' data, two people saving at once.

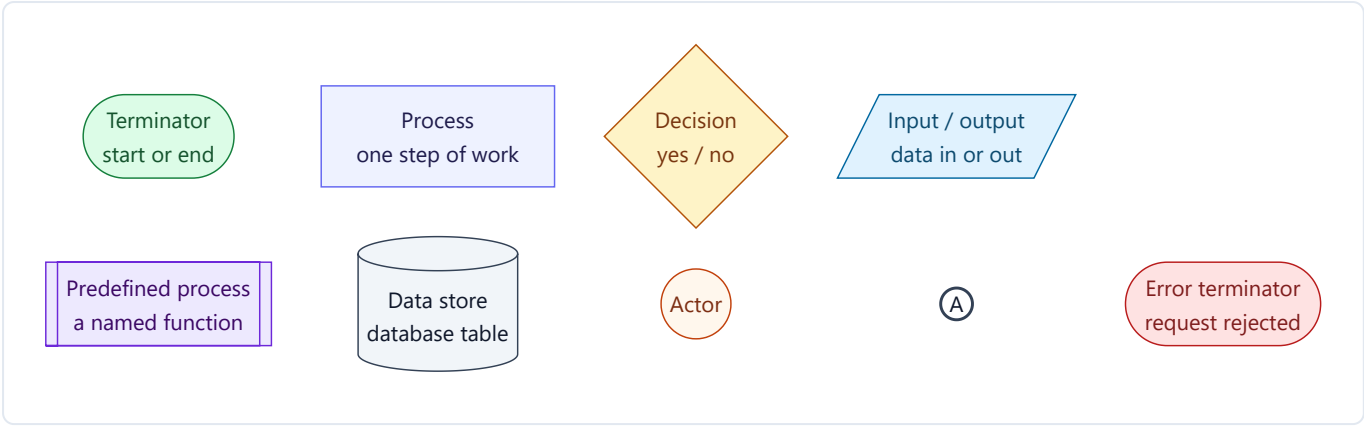
Its audience is the engineers who build, review and maintain the code.

	High-level design	Low-level design
Question	What are the parts and how do they connect?	How does each part work inside?
Detail	Systems, services, hosting, data flow	Modules, functions, SQL, fields, error codes
Typical diagrams	Context, deployment, user journey, pipeline	Flowcharts of logic, sequence diagrams, ERD, API tables
Changes when	The architecture changes	The code changes
In this document	Part 1, figures 1–6	Part 2, figures 7–15

Flowchart symbols

Flowcharts use the standard symbols (ISO 5807 / ANSI style), colour-coded for readability.

Figure 1 · Flowchart symbol legend



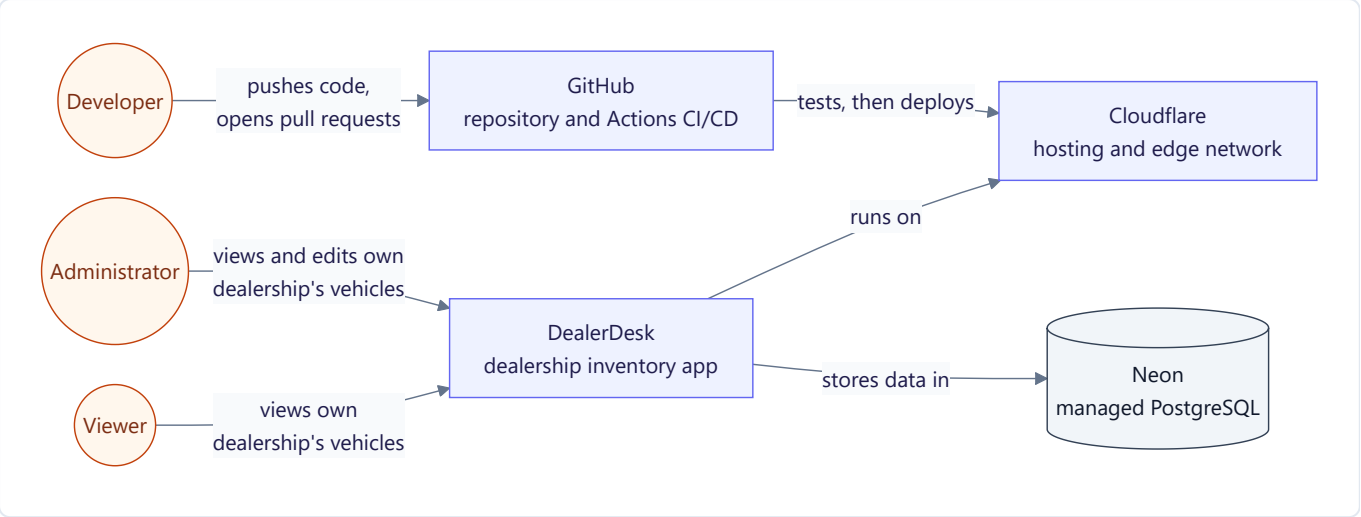
Symbol	Shape	Meaning
Terminator	Rounded oval	Where a flow starts or ends. Red ovals end a flow with an error response.
Process	Rectangle	One step of work
Decision	Diamond	A yes/no question; each exit is labelled
Input / output	Parallelogram	Data entering or leaving: user input, an HTTP response
Predefined process	Rectangle with double sides	A named function described elsewhere
Data store	Cylinder	A database read or write
Actor	Circle with a name	A person or external system
On-page connector	Circle with a letter	The flow continues at the circle with the same letter. Long flowcharts are split into numbered columns this way.

Part 1 · High-level design

1.1 System context

Who uses DealerDesk, and which outside systems it depends on.

Figure 2 · System context

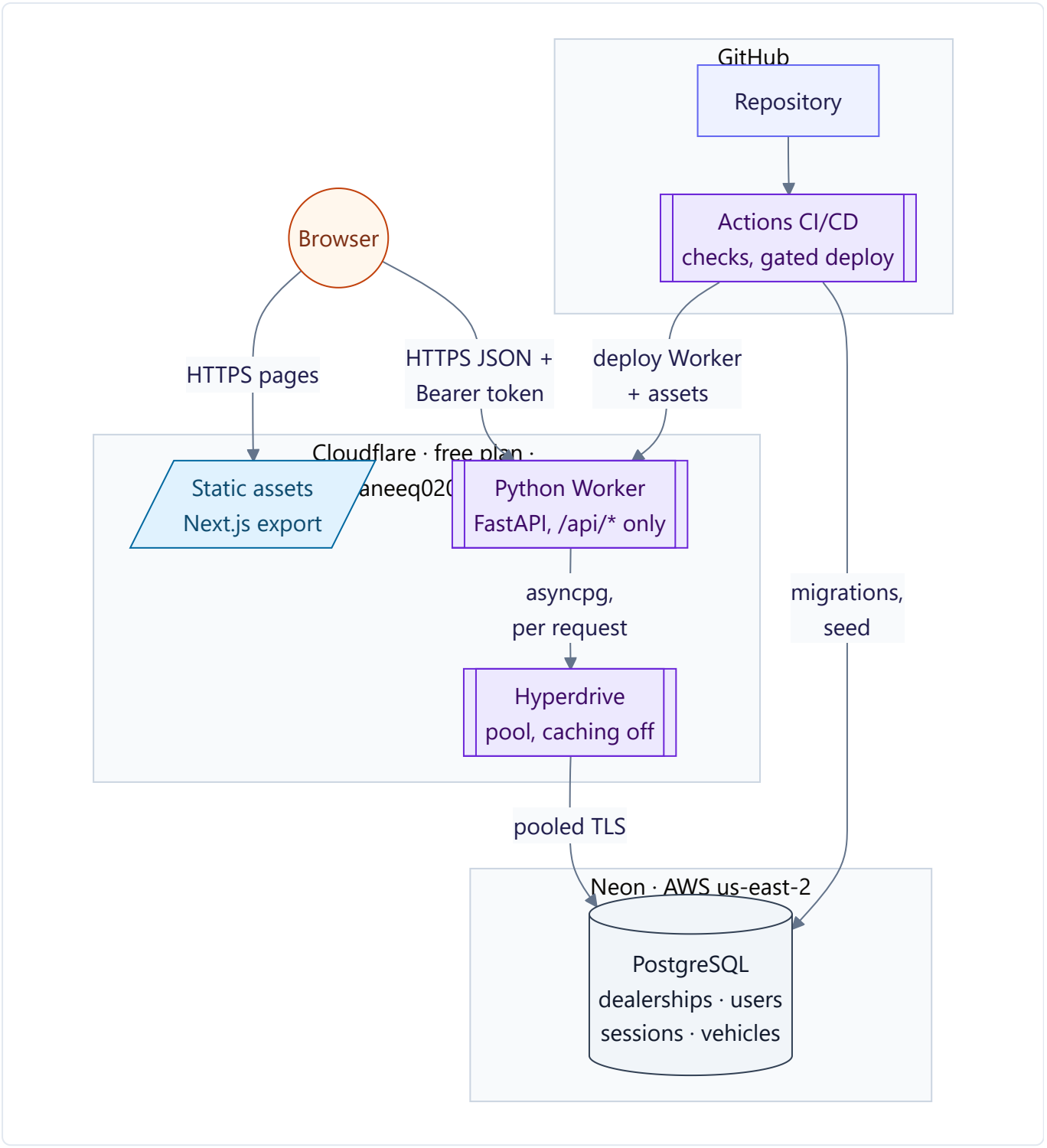


Two dealerships, Northside Motors and Southgate Auto Group, each have two administrators and one viewer. Every user belongs to exactly one dealership and only ever sees that dealership's vehicles.

1.2 Deployment architecture

Where each part runs in production, and what flows between them.

Figure 3 · Deployment architecture

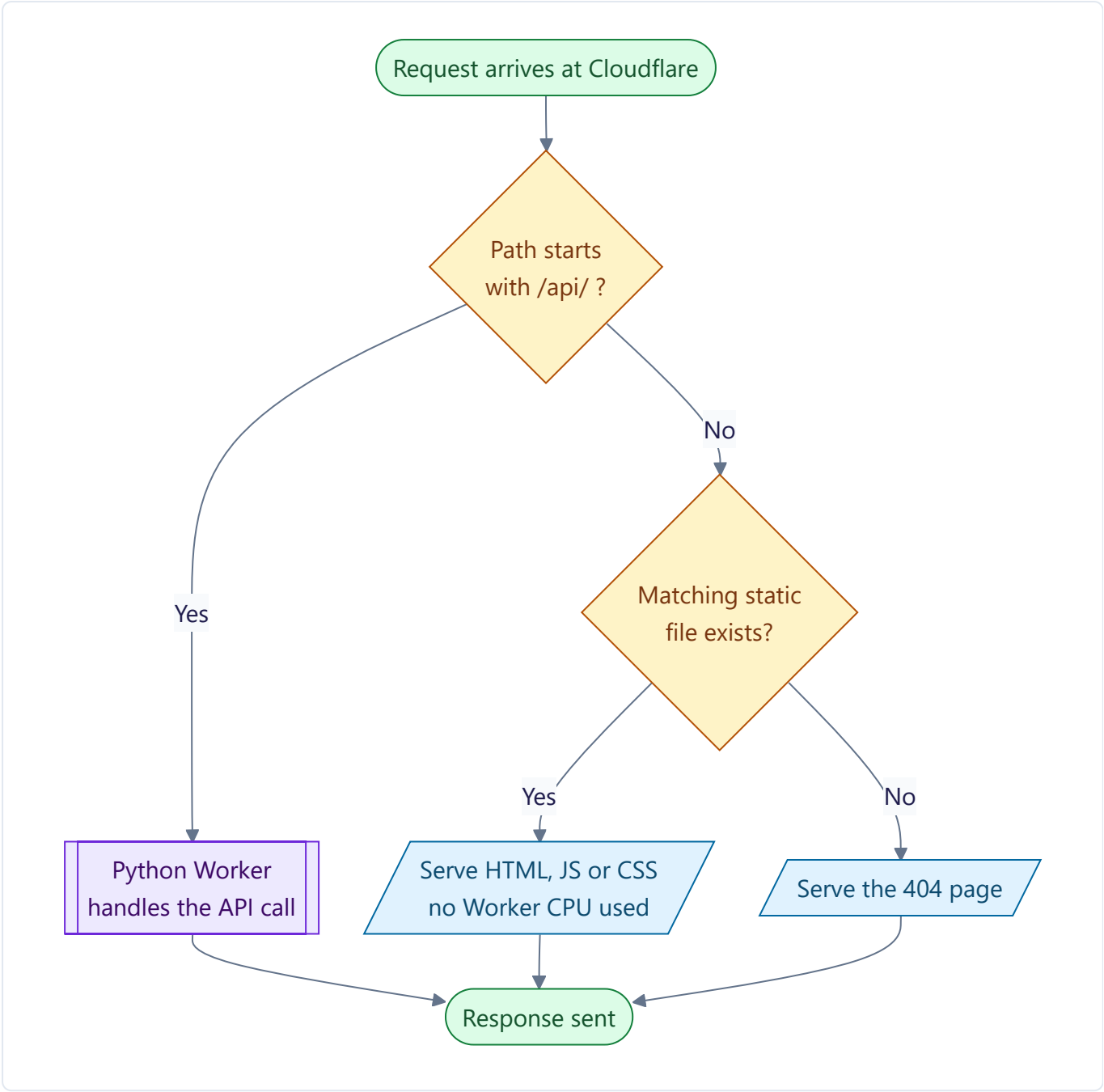


Part	Technology	Why
Frontend	Next.js 16, static export, Tailwind	All data is per-user and fetched with a token, so plain files suffice and cost no compute
API	FastAPI on Cloudflare Python Workers	Python backend required; Cloudflare hosting required
Database	PostgreSQL 16 on Neon, through Hyperdrive	Row locking makes the atomic update correct; Hyperdrive pools connections near Neon
Delivery	GitHub Actions	Tests on every pull request; deploys only from a green `main`

1.3 Request routing at the edge

Every request reaches Cloudflare first. Only API calls run Python.

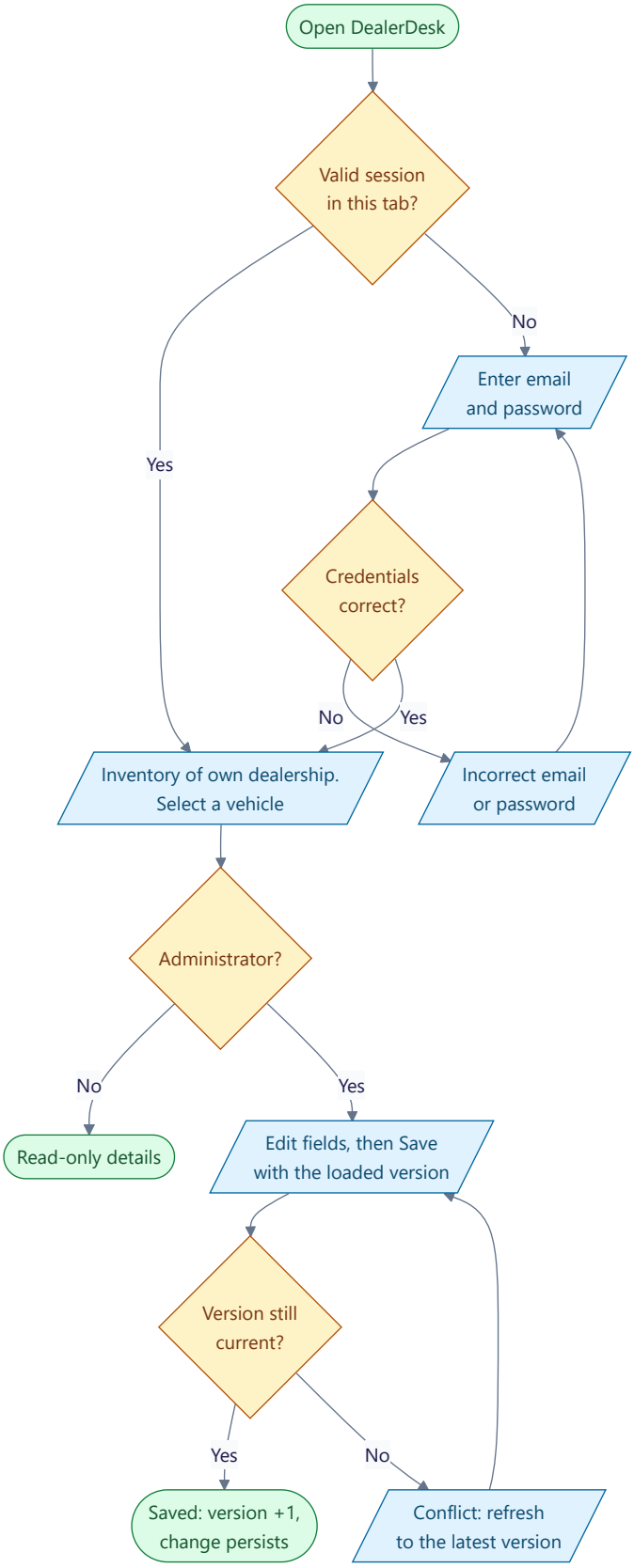
Figure 4 · Request routing



1.4 Main user journey

The business flow from the user's point of view.

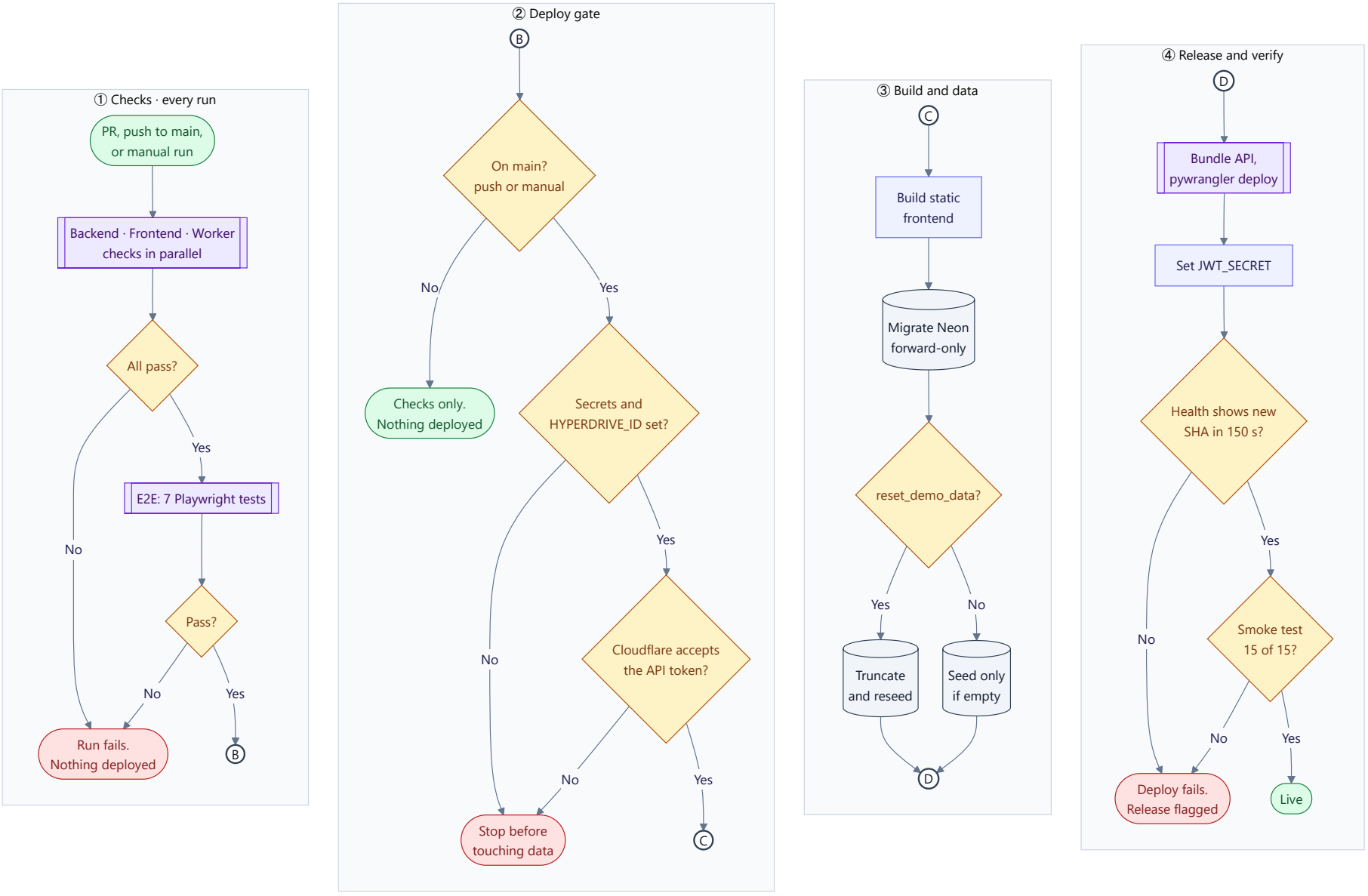
Figure 5 · User journey



1.5 Delivery pipeline

From a code change to a verified release. Each failure stops the run before anything further changes.

Figure 6 · CI/CD pipeline



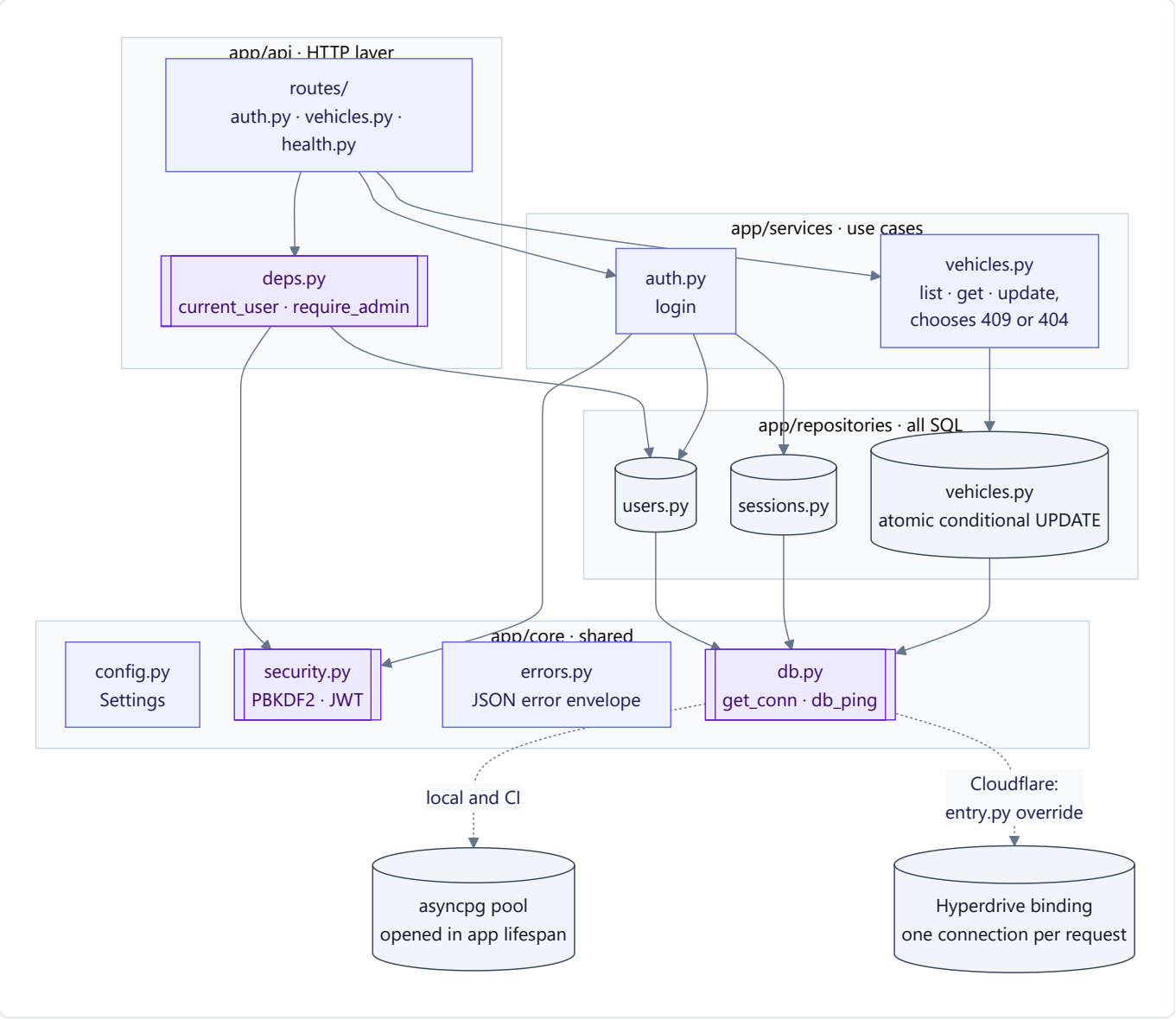
Release recovery. Rerun the workflow manually on `main` with `ref` set to the last good commit, or run `wrangler rollback`. Migrations are forward-only, so a code rollback never needs a schema rollback while migrations stay additive.

Part 2 · Low-level design

2.1 Backend module structure

The FastAPI application is layered. Each layer depends only on the layers below it.

Figure 7 · Backend layers



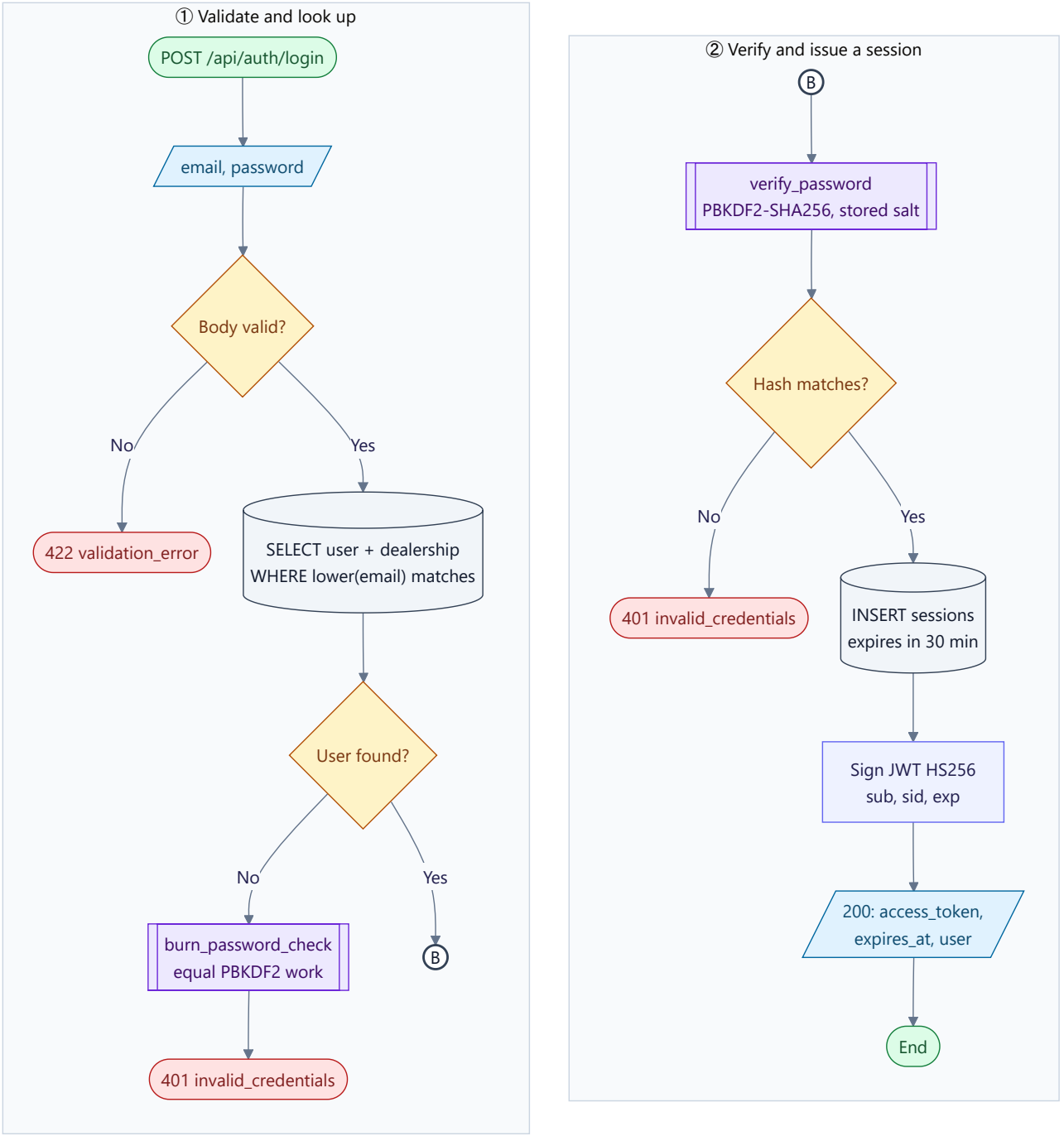
Layer	Responsibility	Rule
`api/routes`	Parse input, pick the auth dependency, call a service	No SQL, no business rules
`api/deps.py`	Authenticate and authorize every protected request	Role and dealership come from the database, never the request
`services`	Use-case logic such as choosing 409 vs 404	No HTTP details
`repositories`	Every SQL statement	Every vehicle query is scoped by `dealership_id`
`core`	Settings, connections, hashing, tokens, error format	No imports from layers above

On Cloudflare, `cloudflare/src/entry.py` builds this same app and overrides three FastAPI dependencies: `get_conn` and `db_ping` use the Hyperdrive binding, and `get_settings` reads the Worker's `env`.

2.2 Login

`POST /api/auth/login` in `services/auth.py`.

Figure 8 · Login flowchart

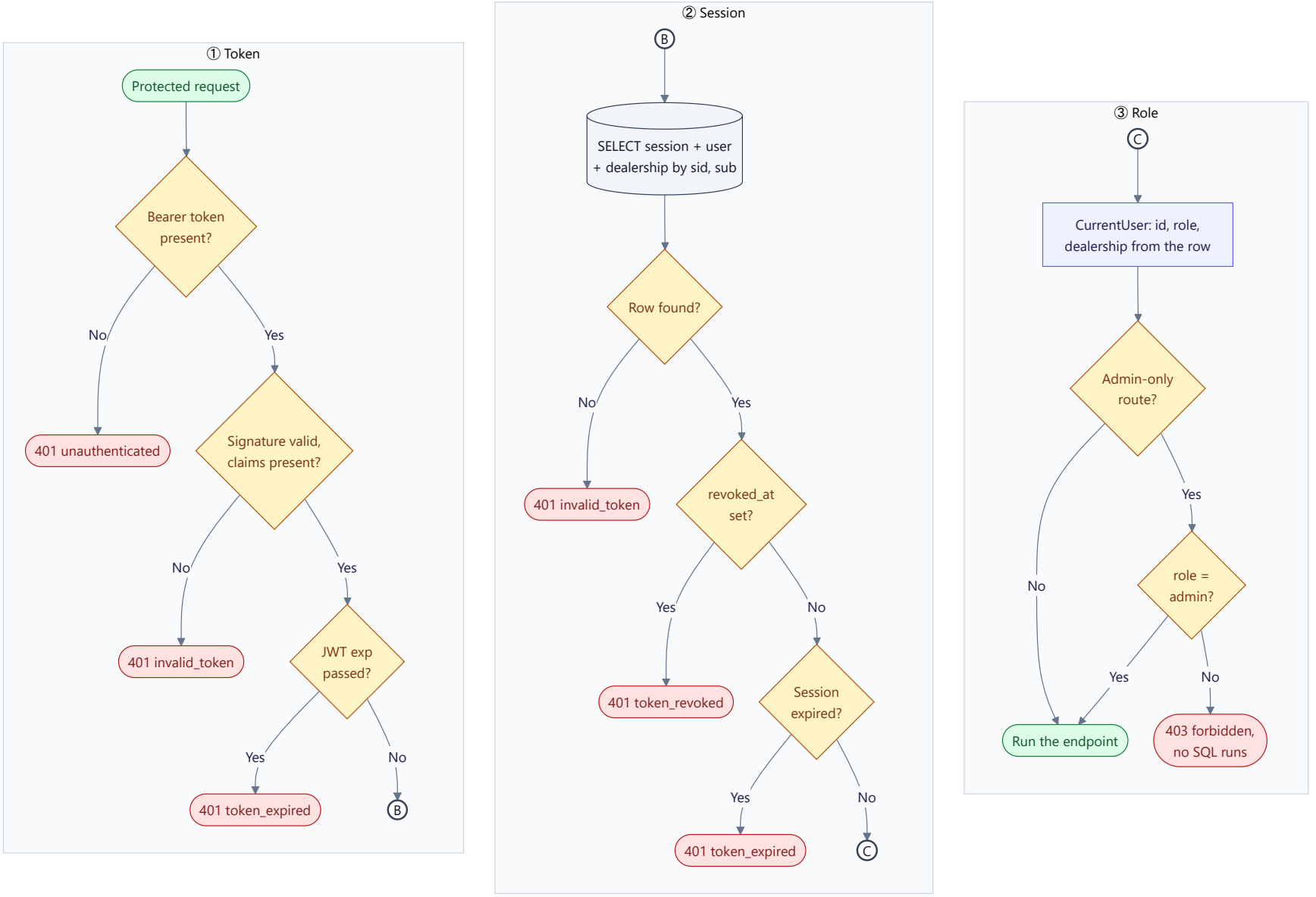


Password storage. `pbkdf2_sha256$<iterations>$<salt>$<hash>`, with a random 16-byte salt per user. On Cloudflare the check runs through the runtime's native WebCrypto; locally it uses `hashlib`. Both produce identical bytes, which is tested against the RFC 7914 vector.

2.3 Authenticating every protected request

`current_user` in `api/deps.py` runs before every protected endpoint.

Figure 9 · Token and session validation flowchart

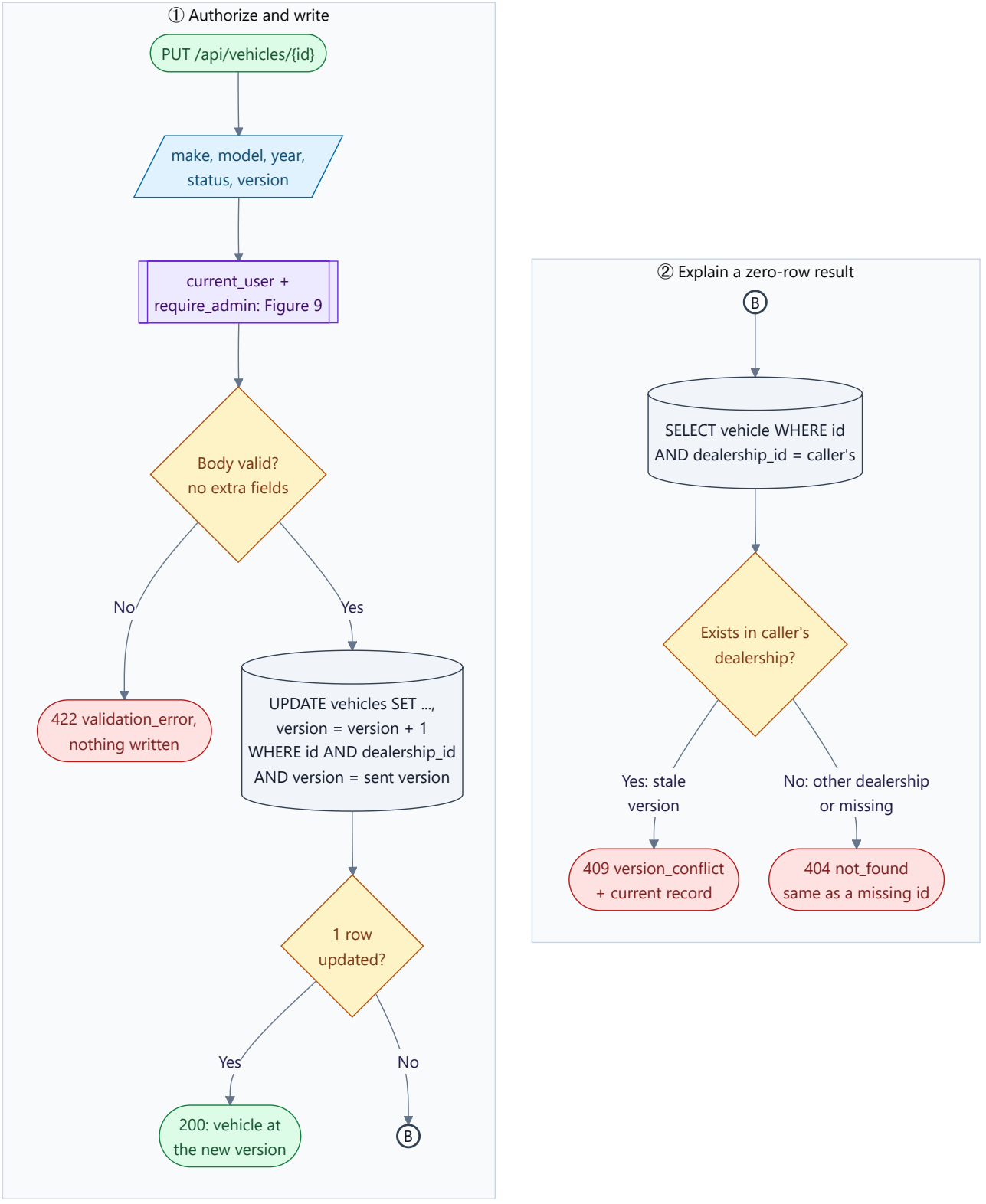


Read the columns left to right; each lettered circle continues at the matching circle in the next column. **Expiry is enforced twice:** by the JWT `exp` claim and by `sessions.expires_at`. **Logout** sets `sessions.revoked_at`, so the same token is refused immediately, even though its JWT has not expired.

2.4 Updating a vehicle

`PUT /api/vehicles/{id}`, admin only. This is the core correctness path.

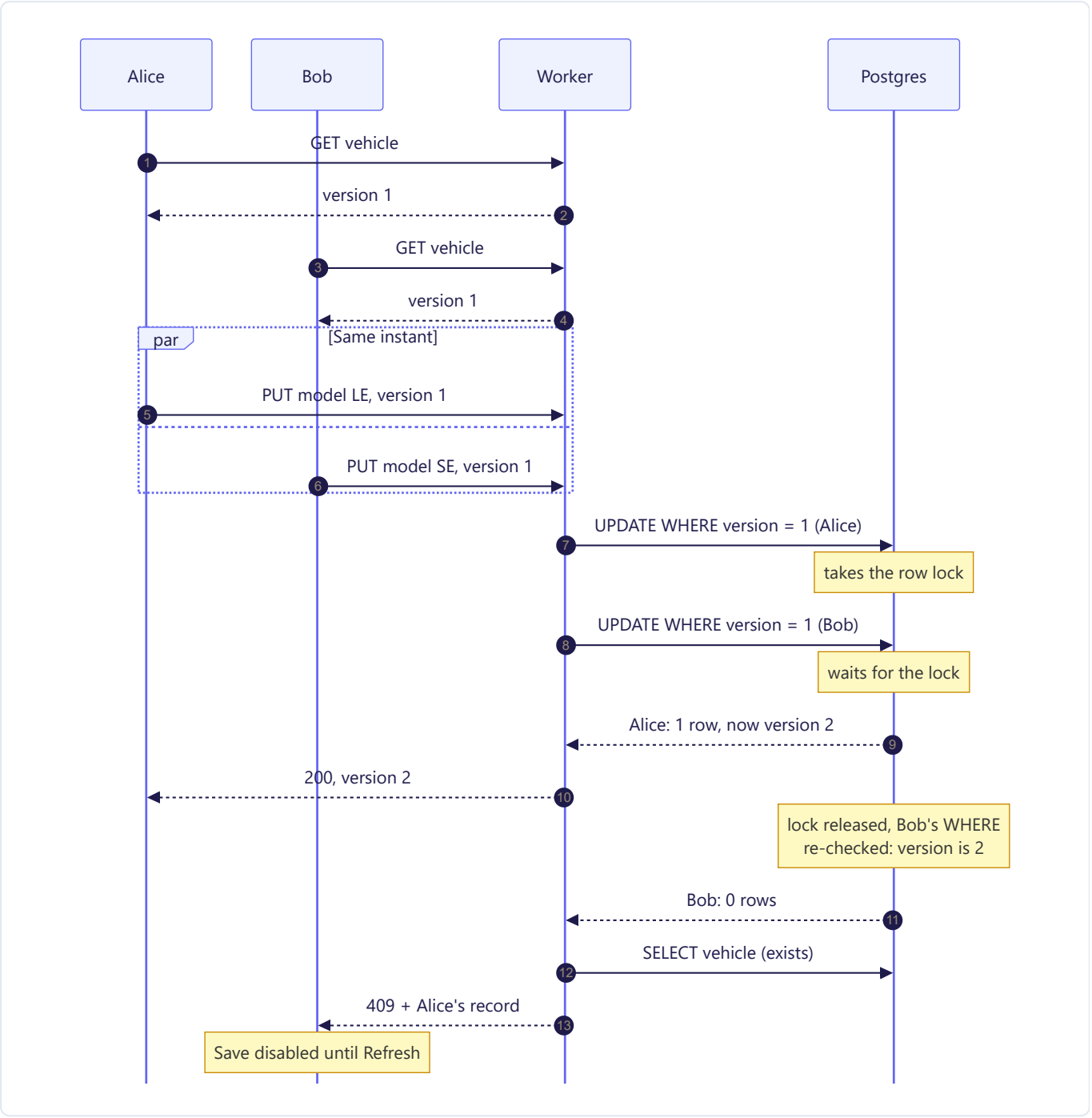
Figure 10 · Vehicle update flowchart



Why the update is atomic. The version check is inside the `UPDATE`'s `WHERE` clause, so checking and writing are one statement with no gap between them. The defective version, commit `309eed5`, read the version first, compared it in Python, then wrote unconditionally. Two requests could both pass the comparison and both write.

2.5 Two administrators saving at once

Figure 11 · Concurrent update sequence

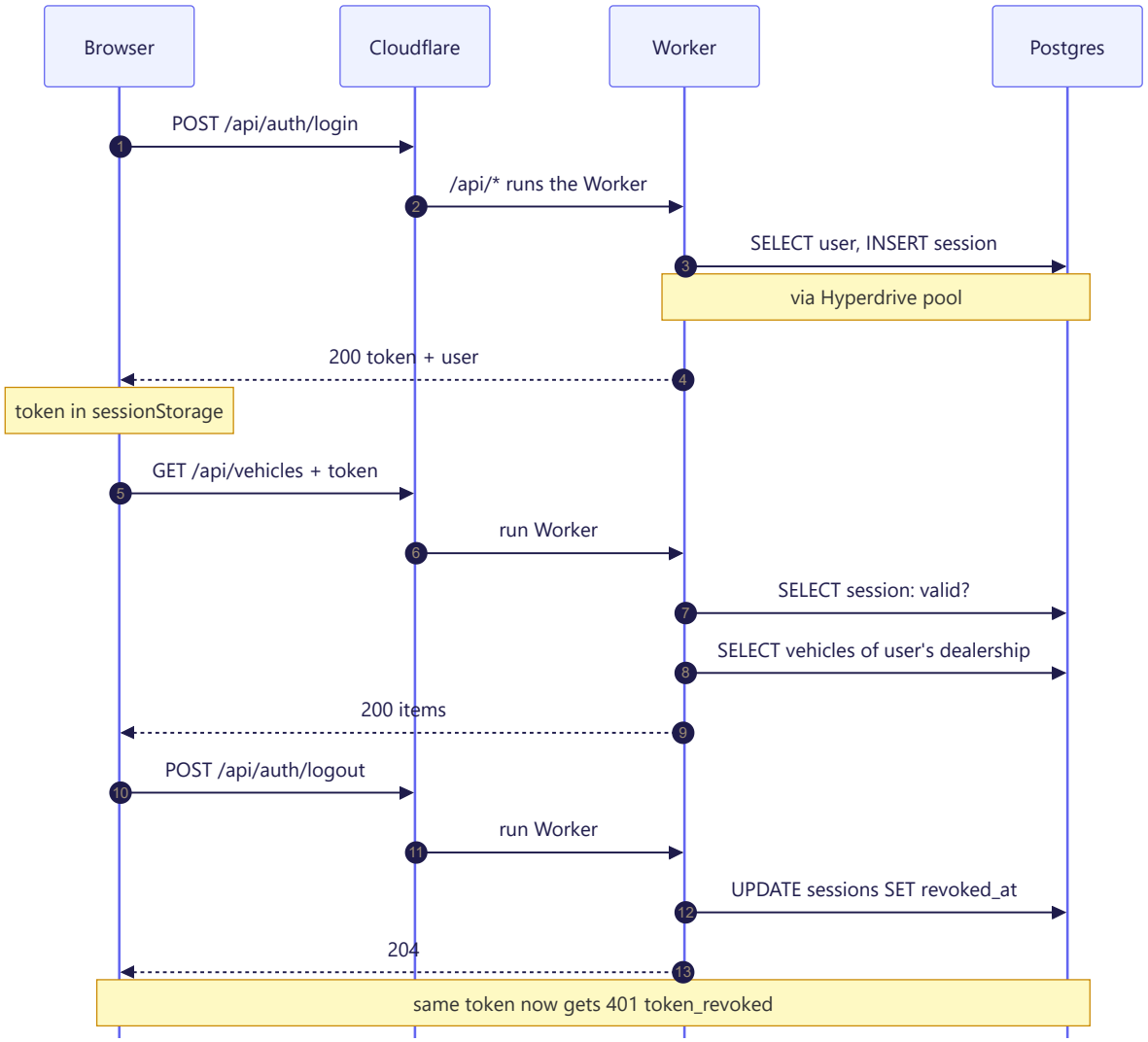


Verified three ways:

Evidence	Result
`tests/test_concurrency.py`, with a barrier forcing the race	Fails on the defective code (`[200, 200]`), passes on the fix
CI	Red run on `de5c4fc`, green run on `6dc1a40`
`scripts/live_race.py` against production	5/5 rounds of truly simultaneous saves, each with exactly one 200 and one 409

2.6 Request lifecycle across the stack

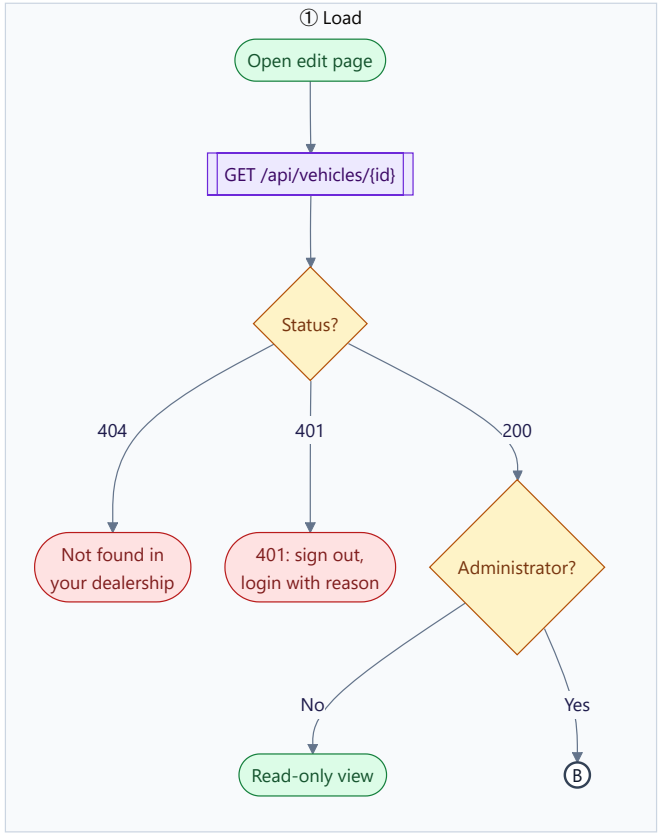
Figure 12 · Login, authenticated call and logout across the stack

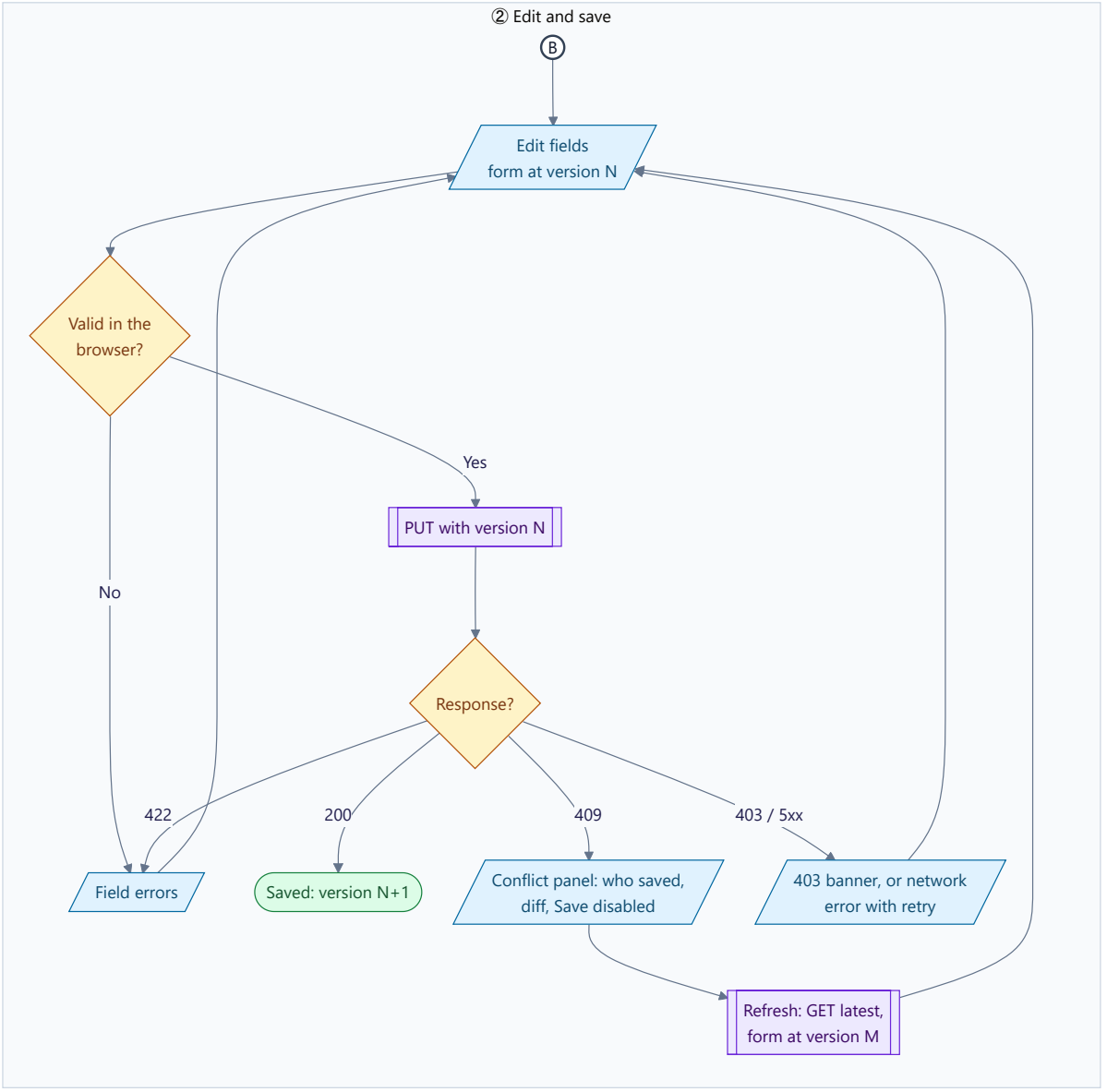


2.7 Frontend: the edit page

`frontend/app/vehicles/edit/page.tsx`, with all API errors mapped centrally in `lib/api.ts`. A 401 at any point signs the user out and returns them to the login page with the reason.

Figure 13 · Edit page flowchart





2.8 Data model

Figure 14 · Entity relationship diagram

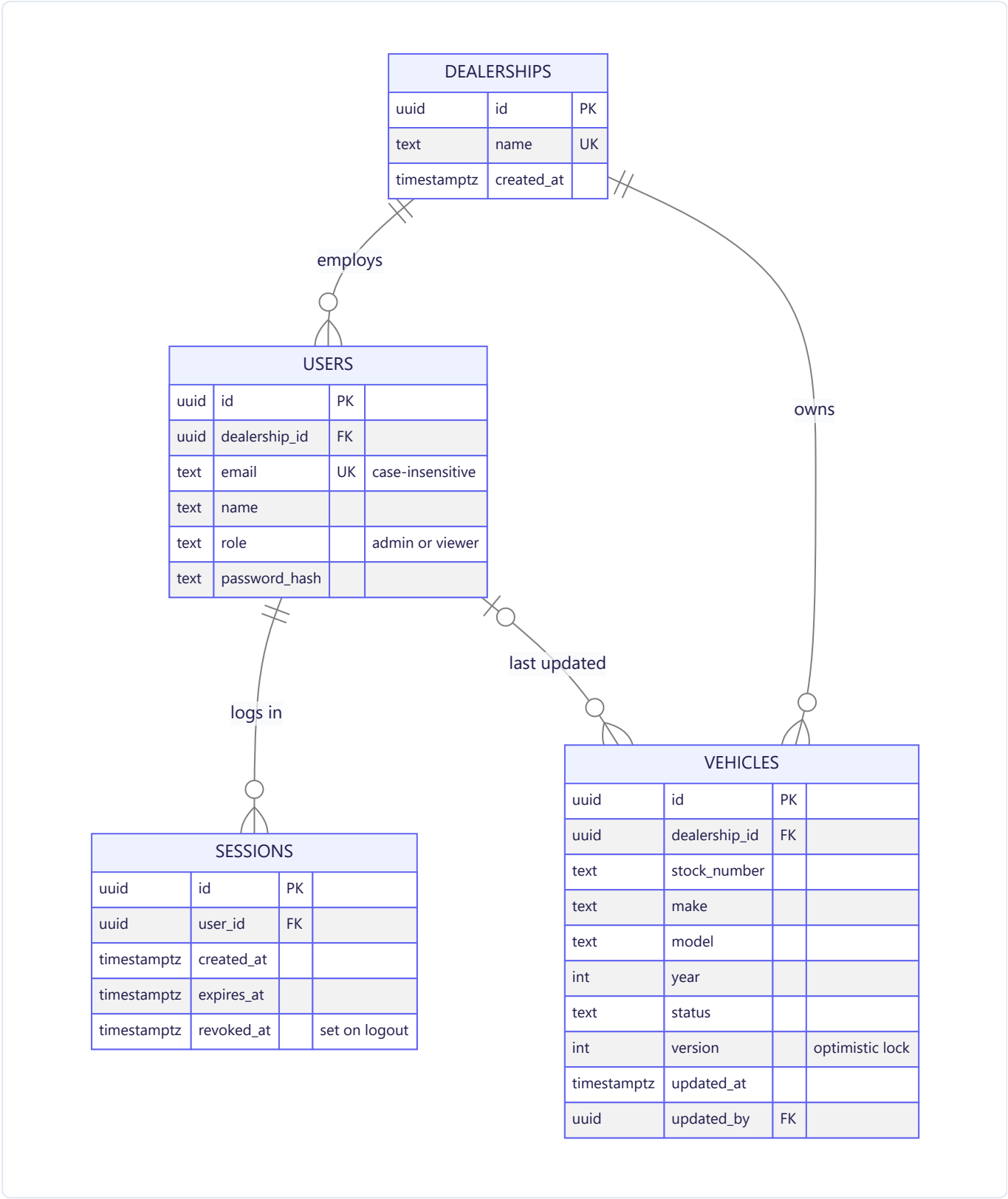


Table	Key constraints
`dealerships`	`name` unique
`users`	`dealership_id` NOT NULL FK (exactly one dealership); unique `lower(email)`; `role` in (`admin`, `viewer`)
`sessions`	FK to user, cascades on delete; `revoked_at` null until logout
`vehicles`	`dealership_id` NOT NULL FK; `year` 1900–2100; `status` in (`available`, `reserved`, `sold`); make and model 1–50 chars; `version` ≥ 1; `(dealership_id, stock_number)` unique; index on `dealership_id`
`schema_migrations`	One row per applied migration file

2.9 API contract

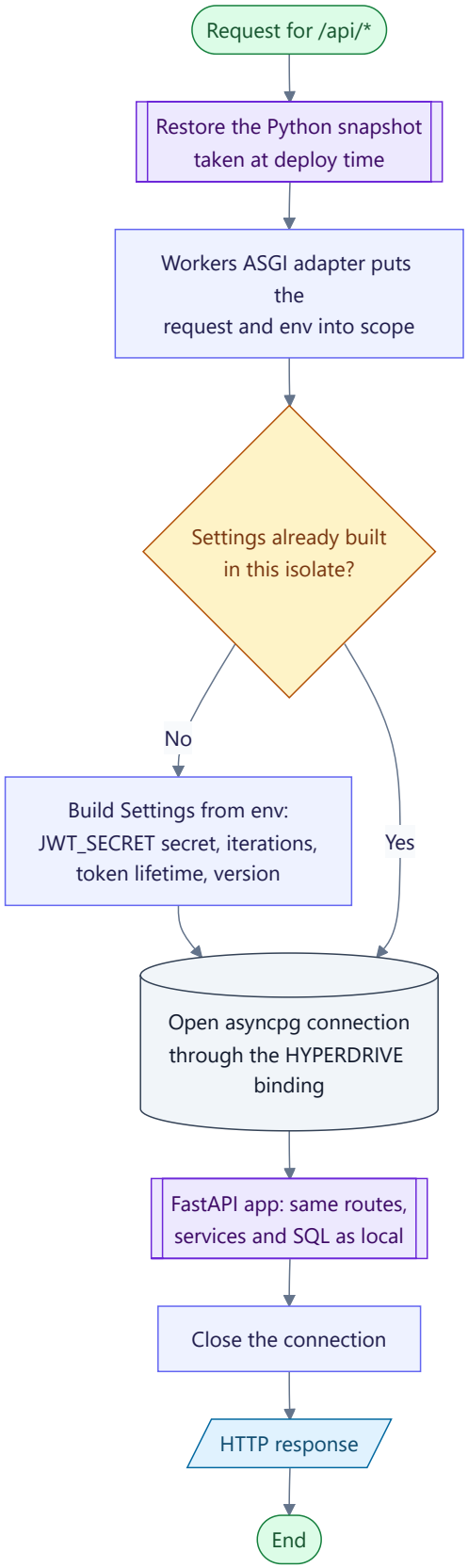
Method and path	Who	Success	Errors
`POST /api/auth/login`	Anyone	200 token + user	401 `invalid_credentials`, 422
`POST /api/auth/logout`	Signed in	204	401
`GET /api/auth/me`	Signed in	200 user + dealership	401
`GET /api/vehicles`	Signed in	200 own dealership's vehicles	401
`GET /api/vehicles/{id}`	Signed in	200 vehicle	401, 404
`PUT /api/vehicles/{id}`	Administrator	200 vehicle, version + 1	401, 403, 404, 409, 422
`GET /api/health`	Anyone	200 `{status, db, version}`	503 when the database is unreachable

Every error uses one envelope, `{"error": {"code", "message", "details?"}}`, so the frontend switches on `code`:

Code	HTTP	Meaning
`unauthenticated`	401	No bearer token
`invalid_token`	401	Bad signature, malformed token, or unknown session
`token_expired`	401	JWT `exp` or session `expires_at` has passed
`token_revoked`	401	Session was logged out
`invalid_credentials`	401	Wrong email or password, indistinguishable
`forbidden`	403	Viewer attempted an administrator action
`not_found`	404	Missing, or belongs to another dealership
`version_conflict`	409	Stale `version`; `details.current` holds the saved record
`validation_error`	422	Invalid fields; `details.fields` lists them

2.10 Cloudflare Worker internals

Figure 15 · Worker request handling



Measured cost. Cloudflare records `cpuTimeMs` per request. Medians run from 25 ms (`GET /api/auth/me`) to 130 ms (login), above the free plan's documented 10 ms. All 45 measured requests were served. The floor is the Python-on-WebAssembly runtime, not the application logic.